

THE EFFECT OF LIQUIDITY CONSTRAINTS ON FIRM INVESTMENT BEHAVIOUR: EVIDENCE FROM SOUTH AFRICAN LISTED FIRMS

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THE EFFECT OF LIQUIDITY CONSTRAINTS ON FIRM INVESTMENT BEHAVIOUR: *EVIDENCE FROM SOUTH AFRICAN LISTED FIRMS*.

INTRODUCTION

Past studies have demonstrated that in the financial markets, liquidity appears to be one of the most significant market frictions that impact assets prices. Liquidity is the capability to buy and sell assets quickly with small price effect (Gregoriou 2013). Many studies have been published connecting liquidity constraints to capital market circumstances, including the studies considered in this literature review. Gregoriou (2013) reviews Myers (1977) who contends that larger companies have larger access to outside finance. The author proposes that shareholders can yield a return through growing many positive NPV projects available for them to invest in. Therefore, there will be an increase in sales profits size and liquidity, stock proceeds, and thus the value of the firm. Therefore, regarding the theoretical framework from financial markets, firm size will be positively related to stock liquidity.

According to Audrestsch and Elstone (2002), an important work by Stiglitz and Weiss (1981) is that the tendency for a firm to incur credit constraints is not neutral with respect to firm size. Rather, as a result of adverse assortment in a market with asymmetric information the likelihood of credit constraint tends to systematically escalate as firm size declines. In compiling some of the most pertinent corporate finance challenges of this day the issue of the lack of availability of long-term financing to firms in developing economies has made itself out to be a stand - alone argument; for example, government initiatives to increase access to finance have largely been partial towards established firms overlooking South African small and medium sized enterprises (Rogerson 2004).

In the following review of literature, we critically discuss the underpinning theories and key empirical evidence that is fundamental in developing the discussion that posits the effects of liquidity constraints on investment behavior of South African firms of different sizes, in confident hopes to achieve the following objectives:

- To examine the relation between liquidity constraints and investment behaviour on JSE-listed firms.

- To examine the effect of liquidity constraints on investment behaviors for firms of different sizes.

This literature review will first look at the theoretical framework from relevant studies. We will then review the empirical evidence which contains the summary of studies, limitations of the studies and theoretical implications. We will then draw our conclusions.

THEORETICAL FRAMEWORK

Market efficiency theories

Under the assumption of a perfect capital market, Modigliani and Miller (1958) believed that a firm's financial structure and policy are irrelevant to its investment decisions, in other words that external funds are a costless substitute to internal finance. This assumption however, is not realistic and has been challenged greatly. It is therefore of great importance to shift focus from a perfect capital market to one that is imperfect. These imperfections are brought about by the presence of agency costs and information asymmetry. This therefore nullifies the assumption that external funds are a costless substitute to internal finance.

In relating investment behaviour and firm size to liquidity constraints, Audretsch and Elston (2000) say that in a market with asymmetric information, the probability of credit rationing tends to increase as the size of a firm decreases. This is also asserted by Fazzari, Hubbard and Peterson (1988) who found systematic evidence that liquidity constraints tend to be more binding as firm size decreases. Liquidity constraints, in this regard, are referred to as the financial factors such as the access of capital markets, availability of internal finance, and access to new debt and equity finance that prevent the availability of cash for the near future, after accounting for the financial obligations for that period Muganga (2010).

Larger firms have generally been in business for longer periods than smaller firms and more information on the companies is available to possible investors or credit providers. Audrestsch and Elston (2000) "Larger firms can finance capital expenditures from internal resources, issuance of equity, or debt. By contrast, smaller firms are limited in the extent of their internal earnings and the potential for issuing equity". This has enabled them to build and sustain constructive relationships with financial agents. This therefore increases the pool from which these larger counterparts can source funding from.

As interest rate in the capital market rises, so too does the riskiness attached to the borrower thereby causing capital providers to limit the amount of they lend (Stiglitz and Weiss, 1981). Smaller firms often have more strict conditions when it comes to credit rationing as observed by Peterson and Rajan (1992, Pg. 3) and (Jaff and Rusell, 1976). In other words, since little information is known on the managerial capabilities and investment opportunities, credit providers might attach a higher default risk to the firms.

The Agency Theory and Asymmetric Information

Based on the premise that managers will not always envision the shareholders' interests above their own, The Agency Theory, developed by Jensen and Meckling (1976), show that the Agency Problem arises when interests of the Principal and the interests of the Agent diverge. Dimarzo, Fishman, He and Wang (2012) argue that the firms' problem of acquiring capital stems from the fact that outside investors are privy to the notion that managers' incentives to act in their own interests outweigh the shareholders' interests. Bouckova (2015) states that the concept behind the Agency Theory provides a "pivotal framework" through which the understanding of complex enterprise processes is achieved more so through the perspective of the Principal-agent. Khan, He, Akram and Sarwar (2017) reports that the degree to which investment behaviour is efficient is adversely impacted by the Agency Problem and financial constraints. Therefore, it can be argued that firstly, the mismatch of the shareholders and management's interests derived from the agency problem are largely responsible for exacerbating the liquidity constraint gap that is prevalent in firms of different sizes and secondly, that the firm's investment decisions are anchored to these effects.

Despite this relationship between management and the shareholders, Gwatidzo, Ntuli and Mlilo (2015) report that Asset Tangibility is a pivotal concept as lenders may demand more collateral to reduce the probability of asset substitution. In order to secure lower loan costs firms can go on to pledge these assets as a form of collateral. Lei, Qiu and Wan (2018) find that firms with low asset tangibility tend to face high external financing costs. Therefore, it can be argued that if the size of the firm is directly proportionate to the current value of assets held, larger firms with high asset tangibility would therefore incur lower external financing costs compared to its smaller counterparts.

Fazzari, Hubbard and Petersen (1988) wanted to prove that "financial constraints are a significant determinant of a firms' investment decisions" The Trio expands reporting that asymmetric information makes it near impossible for providers of external finance to evaluate a prospective firm's investment opportunities and that this could be the leading cause as to why

firms of different sizes, particularly small firms, have much difficulty in securing external finance. Angela Motsa & Associates (2004) attributes the inability for external finance providers to evaluate a prospective firm of a particular size's investment opportunities to the fact that these lenders were groomed in an economy that was historically heavily 'large firm – concentrated' where banks did not have the sufficient skills to assess start - ups and small businesses and that they were unable to "interact efficiently" with entrepreneurs due to the "quality of communication and the level of understanding of their businesses with bank officials." Therefore, it can be put forward that smaller and younger firms are largely penalised due to a lack of historical reputation as well as the inherent risks that come with being not fully grown in giant-filled industries. This argument can also be extended so as to explain why, in retrospect, the failure rate of SMME's is so unreasonably high with a previous lacklustre growth and innovation rate.

In corporate finance, the pecking order theory (or pecking order model) by Donaldson (1961) was popularized by Myers and Maljuf (1984), it postulates that the cost of financing increases with asymmetric information. Financing comes from three sources, internal funds, debt and new equity. Companies rank their sources of financing, first preferring internal financing, and then debt, lastly raising equity as a "last resort". Asymmetric information exists when one party has access to superior information when compared to other parties; that is, between managers and shareholders, there exists different views regarding a firm (Sheikh, Shakeel Ahmed et al. 2012). Asymmetric information affects the choice between internal and external financing thus affecting the liquidity of the firm. Therefore, there exists a pecking order for the financing of new projects. Getting external funding becomes more expensive as asymmetric information insiders and outsiders increase.

The assumption drawn from the pecking order theory, is that managers prefer old investors as opposed to new ones due to the rush that comes along with new investments undertaken. Investment behaviour is affected by liquidity constraints of which is the lack of cash for a business to continue with its operations or invest in a new project. If a firm cannot fund its investment's due to finances, it becomes a problem. Some firms are financially constrained because of their firm size and the state of the economy (whether it is a developed or developing country). The Pecking order theory is relevant to the topic being researched because it deals with the cost of internal and external funding in the presence of information asymmetric.

Investment theories

Tobin and Brainard (1977) reported that The Neoclassical Theory of Corporate Investment as developed by Jorgenson (1963) is based on the assumption that management seeks to maximize the current net worth of the company, which is the market value of the common outstanding shares. It is suggested that investment projects should only be undertaken only if the undertaking of the project would cause an increase in the value of the shares. "If the value of the project as appraised by investors exceeds the cost, then the company's shares will appreciate to the benefit of the existing stockholders." (Tobin and Brainard 1977).

The Q theory of investment developed by Keynes (1936) and expanded by Tobin (1969) has many theoretical advantages which endorses the theory's current use today in that it "focuses on the future market valuation of the firm's assets and not on lags of past variables" as Ajide (2017) records in his study in which he uses the abovementioned investment theory to document the determinants of corporate investments in Nigeria. An extension of the Q Theory of Investment, Tobin's Q, a structural model, posits that the rate of investment is a function of Q – the ratio of market value of new additional investment goods to their replacement cost - the value of growth opportunities a particular firm has.

According to John and Muthusamy (2011) the Investment decisions and the financial decisions of an enterprise is interdependent in the sense that the decision of investment is adopted in relation to the source of financing. Khan, He, Akram and Sarwar (2017) refers to investment efficiency as the balance between over and under investment. John and Muthusamy (2011) describes the Overinvestment Theory: "It can be explained as investment expenditure beyond that which is required to maintain assets in place and to finance expected new investment in positive NPV projects where there is a conflict between manager and shareholder." Khan et al (2017) contributes to this argument referring to the overinvestment hypothesis as a situation that arises due to mismatched manager and stakeholder interests causing management to "exploit their discretionary rights" by investing surplus cash flows in projects that do not offer long term value to shareholders. Underinvestment is based on a "Liquidity affect" in that firms with a large amount of debt obligations invest minutely even if the best investment opportunities come about. This is corroborated by Crnigoj and Verbic (2017) findings that Slovenian listed firms heavily exploited credit facilities and sold more assets to fund their operations in fear that banks would restrict future access to finance causing the firms in question to forego lucrative investment opportunities.

Therefore, it is undeniably appropriate to posit that a firm's investment behavior is largely a product of the extent to which liquidity constraints are experienced, the size and historical reputation of the firm, the capital structure and financial policies governing the firm, the presence of skewed information that is an indication of market and institutional imperfections, the availability of finance at the precise time, whether internal or external, and the existence of diverging management-shareholder interests amongst a myriad of other determinants.

EMPIRICAL FRAMEWORK

International Evidence

The theoretical framework discussed the founding theories on relevant literature. Under perfect market conditions Modigliani and Miller (1958) believed that a company's financial structure and policy are unrelated to its investment decisions. This assumption has been greatly challenged by imperfect market theories such as the Agency Costs Theories and the Pecking Order Theory. As the past literature have verified, liquidity appears to be one of the greatest critical market frictions that impact asset prices. There has been a rise of a massive literature examining the relationship between firm size and liquidity. This is mainly because Myers (1977) argues that larger firms have greater access to external finance. This suggests that shareholders will benefit by increasing the pool of positive net present value projects available for them to invest in. This will in turn increase trading volume and liquidity, equity returns, and thus the value of the firm. Therefore, according to the theoretical literature in market microstructure, firm size will be positively related to stock liquidity. This finding is reaffirmed by many studies as follows:

A study by Cleary 1998 examined the sensitivity of company investment decisions to liquidity position utilizing data for 1,317 U.S. firms over the 1988 to 1994 period. Succeeding the basic method of Kaplan and Zingales (1997), firms are categorized according to financial statement variables that are related to their ability to raise external finance. An objective multivariate classification index, similar to Altman's Z factor, is used to determine firm financial status and this status is allowed to vary from one period to the next. This method captures the preferred cross-sectional properties of a large number of firms and successfully categorizes companies that rise or decline dividends 74 percent of the time. Moreover, a bootstrap methodology was utilized to control for the significance levels of practical differences in parameters across different company classes. The difference of the FHP88 regression equation is projected

utilizing fixed company and year effects where investment in capital is a proxy for, current period cash flow to the firm is used a measure of liquidity constraints. The use of previous year's market-to-book ratio to proxies' growth opportunities that tracks the method of KZ. This is different from FHP88 who estimate Tobin's q. Perfect and Wiles (1994) asserts that developments found from the more complex calculation of Tobin's q are restricted.

Regression estimates by Clearly (1998) are attained by utilising OLS and utilizing static firm and year effects. The estimates for the fixed effects were obtained by utilizing two standard methods that alter the real observations before running equations utilizing the converted values. The results suggest that liquidity and market-to-book are significant causes of investment for all specified groups. Big sample evidence shows that the investment decisions of firms with high solvency are significantly extra sensitive to the accessibility of internal funds than are companies that are less solvent.

A study by Fagiolo, and Luzzi (2004) examines whether liquidity constraints impact companies' size and growth dynamics utilizing a big longitudinal sample of Italian manufacturing companies. The Study is based on firm-level observations from the AIDA database. The testing period used is from 1995-2000 because the authors wanted to keep statistical consistency. The final balanced sample consists of 14277 observations (per year). The chief objective was to examine whether any obvious departure from the "Law of Proportionate Effects" can be better clarified by captivating the connection between financial issues and growth. The authors use a simple regression model that contains the following variables: scaled cash flow as a proxy for liquidity, firm age and firm size, the study used annual data on workers (EMP) as the main proxy for firm size. Other proxies of firm size such as sales (SAL) and value-added (VA) are considered as well so that robustness of the can be checked. The study used non-standardized, pooled, values while adjusting for time fixed effects, the industry effects, and a "subsidiary" dummy. The results were that liquidity constraints produce a negative, statistically significant impact on investment and firm size.

A study conducted by Gregoriou (2013) examined the impact of liquidity constraints and investment opportunities for the UK firms. The data was obtained for all LSE listed firms between the periods of 1990-2009. The sample was split into two separate sample (FTSE 100 and the FTSE All Share Index). Total number of firms used were 619. Following the mainstream literature on external financing available to a firm and liquidity (Becker-Blease and Paul, 2006), the author assume a multivariate equation where: capital expenditure is a

proxy for investment opportunities. The author explains that there is high correlation between investment opportunity and the external finance available to a firm. The author create three proxies for the share price liquidity: the illiquidity ratio reputable by Amihud (2002), sales size, and share returns. The study used GMM panel estimator developed by Arellano and Bond (1991) to deal with the indigeneity of the explanatory variables. The results of the study found a positive and significant statistical relationship between investment opportunities and stock market liquidity for both large and small enterprises.

In a conjunction with the above study, we review a study conducted by Masso (2002) who studied if there is underinvestment relating to financing constraints in Estonian manufacturing firms. Firm level panel data was used from a period from1995 through 1998 with several items from financial statements used. A final sample of 373 firms in manufacturing industries was considered. The entire firms in the database was in this period was 4500, of which 373 firms in manufacturing industry. The study considered Survivorship bias, Firms with negative or zero fixed tangible assets (excluded) and firms with super natural growth (5%upper and lower tales of distribution) to control for outliers (scale effect) were also excluded. The existence of liquidity constraints was tested in two ways:

Primary, reduced form equations were estimated in order to detect whether internal finance impacts investment. The following is included in the equation: Cash flow and cash stock as liquidity variables, the liquidity variables proxy for internal net worth, while they also carry information about what amount of investment spending can be internally Total debt was also included in the equation. The intuition for including the leverage variable is that agency costs occurring due to diverging interest of lenders and borrowers (are assumed to increase in the amount of debt used. The current firms in the sample are not listed on the stock market so the Author was unable to calculate Tobin's Q, so instead employment growth was used to control for the existence of investment opportunities, as with Bratkowski et al (2000). Firm size is used as a proxy for the capability to get funding through external funds. The sample is divided into three equally sized groups, conferring to the average size of tangible assets over the sample period.

Secondly, the Euler equation is used. The paper used a model that was estimated by Bond and Meghir (1994), advanced by Bond at al (1998). The variables included in the equation are cash flow, output net of adjustment costs, the user cost of capital and investment. Lagged investment terms consider the effect of adjustment costs. The estimated equation controlling for the

relation between current profits and expected future profitability (Gaston and Gelos, 1999). The study used the GMM estimator which is consistent on estimating the coefficients of the specified model (Bond and Mehgir, 1994). The study found from both simple OLS regressions and the Euler equation that small (and Estonian) firms are more dependent on their own cash flow and cash stock than larger firms. The results provided evidence of liquidity constraints presence and that cash flow affected investments.

The study by Hericourt and Ponket (2009) examines the success of the continuing financial system improvements in China by examining the degree to which companies are financially constrained. The authors emphasis is on the impact that Foreign Direct Investment (FDI) have in funding the Chinese corporate sector, and the analyses is done to examine whether incoming foreign investment in China plays an important role in alleviating domestic firms' credit constraints. The study used firm-level data on 1300 domestic companies over the period 2000-2002. First, the paper shed light on the impact of the ongoing reforms of the financial sector by using a structural model to assess the importance of credit constraints in China. Second, the authors focus on the part played by FDI in funding the Chinese corporate sector, this is done by asking whether FDI eases or exacerbates domestic firms' credit constraints. Following a model by Bond and Meghir (1994), the authors estimate a dynamic investment model and estimate an equation with the following variables: cash flow to for liquidity, investment I fixed assets, real output and real user cost of capital. To test for different ownership effect, the sample is split between private and state-owned companies. The study find that two firm-level measures of financial distress do significantly affect investment for domestic private firms, suggesting that they are credit constrained.

A study by Wang (2014) investigates how finance affects corporate investments and profitability. The focus was on the Asian markets from 2005 to 2011, non-financial firms from Indonesia, Malaysia, The Philippines, Singapore and Thailand. The Tobin's Q was employed to capture growth opportunities for firms in each of these economies. Evidence from the study show that revenue has a greater power of explaining investments which highlights the fact that Tobin q doesn't capture all of the growth opportunities (Whited, 1998). The overall results from the paper suggest that financial conditions do affect the firms' investment behavior which thus affects growth.

A study conducted by Cull, Li, Sun, and Xu (2015) investigates the connections between firms' government relations, investment behavior, and financial frictions for a sample of 200 firms

from the World Bank survey. The survey was conducted for a total of 12400 Chinese manufacturing companies from 120 towns back in year 2005. Following the FHP methodology, the study assumes a parsimonious model of investment with the following variables: investment, capital stock, a course of covariates taking basic firm and regional features, external finance, and pervious period capital stock, entrance to bank funding, allowance of credit, cash flows, and sales. For growth opportunities, both sales growth and firm level Tobin's Q ratios were used. For estimation technique the cross sectional estimation method and fixed effects model was used. Many of the variables had outliers, such as investment intensity sales intensity, CF, and sales growth. The authors therefore winsorize the observations using a 1 percent tail wherever appropriate to reduce the influence of those outliers. The results of the study concerning liquidity constraints on investment was the data therefore show that minor companies, which have high growth opportunities, are more probable to experience relatively more financial constraints. The study also assumed a pooled model outcomes without accounting for growth opportunities or proxies for excess to credit. The pooled results demonstrate that on average, Chinese companies face financial difficulties because their rates in their investment are sensitive to excess of bank funding, and are also sensitive to cash flows under the hypotheses that the FHP technique stay valid.

The results based on the pooled sample suggest that on average Chinese firms face financial constraints since their investment rates are sensitive to access to bank loans, trade credit, and to the availability of collateral assets, and are sensitive to cash flows under the working hypothesis that the FHP method remains valid in the Chinese institutional context.

(Mulier, Schoors et al. 2016) investigated the financial constraints and the investment-cash flow sensitivity relationship for European Small, Medium Enterprises (SMEs). Investment of firms with low dividend payout ratios are sensitive to the availability of cash flow (Fazzari, Hubbard and Peterson, 1988), this paper found that financially constrained firms have a higher cost if debt, compared to the more financially flexible-larger firms. Mulier et al (2017) constructed an index for the age-size-cash flow-leverage to find how cash-flow sensitive firms are in terms of investment. According to this index, Mulier et al (2017) found that financial institutions prefer larger firms due to the sufficient information available and older firms have a proven credit record as opposed to the young, small firms. Results from the index show that firms that are financially constrained have the highest investment – cash flow sensitivities.

A study done by Gul, and Shahid, S. (2017) examines the relationship between the investment opportunities and liquidity constraints in the two South Asian emerging markets, (India and Pakistan) over the period of 2010–2015. The study examines non-financial firms listed on the BSE and the KSE for the specified period. The study had a final dataset of non-financial firms for 208 firms listed on BSE and 136 firms listed on KSE. Whereas, the initial sample consisted of 250 firms in each of the BSE and KSE, but 156 companies was dropped due to missing data and delisted firms from the stock exchange during the study time period. The authors mention a few limitations in the data collection, such as some of the variables information for a particular year is missing in the DataStream database. Such firms for the data are missing in a particular year are removed from the final sample. The authors followed a model by Becker-Blease and Paul (2006) and it had the following variables: capital expenditure as a proxy for investment opportunities, market to book ratio, leverage ratio of the firm external liquidity of the firm, size of the firm and the dividend pay-out ratio. The Using pooled OLS fixed effect model, and the study found a significant negative association between assets illiquidity (external liquidity constraint) and investment opportunities in both BSE and KSE firms. To control the indigeneity between firm internal and external liquidity and investment opportunity, the study employed the GMM estimator system developed by Blundell and Bond (1998). In this methodology, the lagged instruments are employed on the variables to control for indigeneity of the explanatory variables used in the pooled panel data. The findings of this study are consistent with the findings of previous literature that liquidity is positively related to firm's investment opportunities. Our study further clarify that both internal and external liquidity are important for firm's investment opportunities. This finding contradicts the M & M theory that capital structure is irrelevant.

The uneven effect of financial constraints study done by (Lashitew 2017) for the Ethiopian manufacturing industry made use of VAR time series method, which provides a significant relationship between investment and its determinants (Gilchrist, 1998). The results from impulse responses from the VAR estimation shows that cash flows have greater effect on the investment among small firms. This study is consistent with the assumption made by Myers (1977) of asymmetric information, the imperfections in the financial markets undermines the performance of small firms in Africa.

A study done by Ho, Christodoulou, and Prokhorov (2018) builds upon research in the financial constraints literature by examining its one-sided effect on the frictionless level of capital investment. Data was obtained from the Compustat fundamentals annual database from 1970-

2016 for the entire pooled sample. The model specification used builds upon the solution to the firm's optimization problem by Wang (2003) where: Tobin's Q and sales are used to proxy for investment opportunity, cash flow used to proxy for liquidity, and capital stock. Using a fixed effects model, the author's results suggest that while the effect of cash flows in mitigating financial constraints has indeed dampened over time, their role in alleviating financial constraints continue to remain relevant.

The case of Bulgaria was investigated by Budina, Garretsen and de Jong, 2018. The paper investigates liquidity constraints on investment performance. The study is based on the choice between internal and external funds, claiming that external finance premium forces firms to prefer internal funds (Myers, 1984) which is inconsistent with the irrelevance theory by Modigliani and Miller (1958) in a perfect capital markets. The case is based on a period 1993-1995, results reveal that Bulgarian firms are liquidity constrained and the access to external markets is due to the failure of the financial system in Bulgaria (1996-1997).

Kumar and Ranjani (2018) conducted a study to analyse how cash flow sensitivity affects the investment activities of firms. The study was done for manufacturing firms listed on the Indian exchange for the period of 2009 – 2015. Using GMM estimator to find the sensitiveness of cash flows to investment, Kumar et al (2018) found that group affiliated firms are less restricted in terms of cash flow as a proxy for financial constraints. They found that market size of firms shows how firms with a higher market capitalisation are able to access the market for external finance than those firms with a small market capitalisation. The study concluded that retained earnings can support constrained firms in order to take on profitable investments.

Evidence from Africa

A study done by (Makina and Wale 2016) on manufacturing firms in South Africa found that asymmetric information has positive investment cash flow sensitivity better than agency costs. The study used the GMM method to test and compare the contributing factors to liquidity constraints. The choice of manufacturing firms was due to the smooth running of schedules compared to other sectors. Using firm level financial data for the period of 2005-2014 for JSE listed firms, the study found that South African manufacturing firms fall victim of asymmetric information, this is illustrated by a high Tobin's Q ratio. The study also made use of econometric tools to test the relationship between cash flow which was used as a proxy for

financing constraints and investment; results from the Wald test suggest that cash flow explains the investment behavior of the firms

The notion of Miller and Modigliani (1958) of financial factors being irrelevant has been greatly challenged, evidence from literature suggest that in an imperfect capital market, internal and external capital are not perfect substitutes (Schiantarelli, 1998; Bhagat, 2013). A study by (Ajide 2017) on firm-specific and institutional determinants of corporate investments in Nigeria (Ajide 2017) reported that firm-specific factors such as cash flows, does have an influence on corporate investment. The study made use of panel data analysis on an 11 – year period (2002- 2012) for 54 non-financial firms. Panel data allows researchers to have a better view of adjustments brought on by the presence of lagged dependent variables in the regressors. The study employs GMM estimator (Arellano and Bond, 1991), political stability, control of corruption have no significant effect on the profitability of firms in Nigeria.

These findings draw two main conclusions: (1) that cash flows as a proxy for liquidity constraints have a negative impact on investment behavior. (2) That smaller firms are more liquidity constrained then large firms because of information symmetries and that it therefore becomes hard for these small firms to excess external funding. These results suggest that the market is not perfect.

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